

Index	Load (N) or x	Extension (mm) or y	x*y	x^2
1	5	1.2	6	25
2	10	2.5	25	100
3	15	3.7	55.5	225
4	20	4.9	98	400
5	25	6.1	152.5	625
Sum:	75	18.4	337	1375
	Sources:			
	https://drive.google.com/open?id=1UMryFJnCQEJW16XyEe_f8uwoJJi8P3Vj			
Linear Regression Slope Calculation			Linear Regression Intercept Calculation	
Formula: $m =$	$\frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$		Formula: $c =$	$\frac{\sum y - m(\sum x)}{n}$
Substituted: $m =$	$\frac{(5 \times 337) - (75 \times 18.4)}{(5 \times 1375) - 75^2}$		Substituted: $c =$	$\frac{18.4 - 0.244 \times 75}{5}$
Final Answer: $m = 0.244$			Final Answer: $c = 0.02$	
Linear Regression Line Equation			When the load is 55 N : ie. $x=55$	
Formula: $y = mx + c$			Extension(y) = $0.244x + 0.02$	
Final Equation: $y = 0.244x + 0.02$			$y = 13.44$	